

Lecture 8: genes, genomes, evolution

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Lecture 8: information, genes, evolution

2 J. Craig Ventner

You might have heard that [Craig Ventner died last Wednesday](#).

He was scientist and technologist – a major contributor to DNA sequencing, the human genome project, the field of synthetic biology (synthetic chromosomes), and more. Fitting that we talk about genes and genomes today.

3 Comparing Genomes: *H. sapiens* vs. *E. coli*

How many times larger is the human genome than *E. coli*? How many more genes does it contain?

	Genome size	Gene number
a.	$\times 5000$	a. $\times 5000$
b.	$\times 500$	b. $\times 500$
c.	$\times 50$	c. $\times 50$
d.	$\times 5$	d. $\times 5$

4.6 Mb ≈ 5 Mb for *E. coli* K12 versus 3.055 Gb ≈ 3 Gb
 $\Rightarrow$ **option (b)**, $\times 500$

4,225 $\approx 4,000$ genes for *E. coli* versus 19,890 $\approx 20,000$ protein-coding genes
 $\Rightarrow$ **(d)**, $\times 5$

Note: the 19,890 number comes from a 2022 publication of a complete genome from a single person. See [Nurk et al. Science 2022](#).

4 Scaling of bacterial genome sizes

Who is working on *Vibrio cholerae*? (None — good!) The genome of *Vibrio cholerae* is 4.03 Mbp. How many genes does it encode?

(a) < 3600 (b) 3600–4400 (c) > 4400

4.1 Scaling of gene number for bacteria

In bacteria, gene count scales linearly with genome size according to the simple rule: number of genes $\approx$ genome size / 1000 bp. This scaling works because

- a typical protein is ≈ 300 aminos ≈ 1000 nucleotides.
- bacteria have compact genomes with little inter-genic sequence.

Applying this rule to *E. coli* lab strain **K12**:

- 4.6 megabase genome $\Rightarrow 4.6 \times 10^6$ bp.
- 4.6×10^6 bp / 1000 ≈ 4600 genes.
- $\approx 7\%$ greater than the accepted value of 4,312 protein-coding genes.

Note: empirical numbers are from the [EcoCyc Database](#). See [Karp et al. EcoSal Plus 2025](#). The accepted value is 4,225. The same rule fails catastrophically for eukaryotes, but within the bacterial domain it stands as a clean quantitative regularity with essentially no free parameters.

Outline for whiteboard

- Comment on HW #2 (?)
- Craig Ventner
- *E. coli* vs Human genome
- genome scaling
- genes = model
- genome sizes $\rightarrow$ repl. time
- mutations coli
- covid α - micro v. macro
- translation errors (skip?)
- specific mutation coli
- human mutations
- 2 mutations

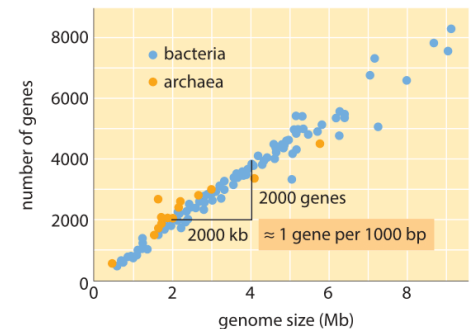


Figure 1: Scaling of gene count with genome size for bacteria and Archaea. From Cell Biology BTN.

	Organism	# of protein-coding genes	# of genes naïve estimate: (genome size /1000)	BNID
viruses	HIV 1	9	10	105769
	Influenza A virus	10-11	14	105767
	Bacteriophage λ	66	49	105770
	Epstein Barr virus	80	170	103246
prokaryotes	<i>Buchnera sp.</i>	610	640	105757
	<i>T. maritima</i>	1,900	1,900	105766
	<i>S. aureus</i>	2,700	2,900	105500
	<i>V. cholerae</i>	3,900	4,000	105760
	<i>B. subtilis</i>	4,400	4,200	111448
	<i>E. coli</i>	4,300	4,600	105443
eukaryotes	<i>S. cerevisiae</i>	6,600	12,000	105444
	<i>C. elegans</i>	20,000	100,000	101364
	<i>A. thaliana</i>	27,000	140,000	111380
	<i>D. melanogaster</i>	14,000	140,000	111379
	<i>F. rubripes</i>	19,000	400,000	111375
	<i>Z. mays</i>	33,000	2,300,000	110565
	<i>M. musculus</i>	20,000	2,800,000	100308
	<i>H. sapiens</i>	21,000	3,200,000	100399, 111378
	<i>T. aestivum</i> (hexaploid)	95,000	16,800,000	105448, 102713

Figure 2: Examples of the naive genome scaling ($\#$ protein coding genes = genome size / 1000) across the tree of life. From Cell Biology BTN.

Caveat: in evolution, generalizations are limited.

5 GeneSweep; genes as model

As the human genome project was going on, there was a contest ([GeneSweep](#)) to guess the number of genes in the genome.

5.1 How did they know where the genes were?

What do you think? In the Human Genome Project, the approaches included

1. Expressed Sequence Tags ([EST](#)); a primitive form of RNA seq
2. Conservation of sequence regions
3. Homology to known proteins (e.g., in yeast or flies)
4. Computational models of genes

Today it's a complex mixture of approaches:

1. statistical models (today AI)
2. RNA-seq, ribosome profiling
3. proteomics
4. genetics + biochemistry experiments
5. homology to other species (e.g., human to yeast)

But when we confront a new genome, we are **applying a model to guess which regions are transcribed and translated**.

This is especially so in animal genomes, which have complex organization including multiple chromosomes with important non-coding elements like centromeres,

Current Genesweep Votes

Genesweep	
Bets	228
Mean	62,598
Median	53,700
Lowest	27,462
Highest	200,000

Figure 3: Typical guesses for the number of protein-coding genes in the GeneSweep contest as of May, 2000.

telomeres, enhancers, introns, etc.

5.2 EGS, Shotgun sequencing, and Craig Ventner

This [post by Steven Salzberg](#) reviews J. Craig Ventner's contribution to the human genome project, specifically focused on "shotgun sequencing." This is a randomized approach very similar to ESTs mentioned above – you sequence small pieces and assemble them into genomes computationally afterwards. Similar to Illumina sequencing.

6 Reconciling the 500-fold difference between *E. coli* and human genome size

We talked about proteins being about 300 amino acids. Typical human genes are a bit longer (≈ 375 AA, genome weighted average).¹

$5\times$ more genes and 25% larger proteins **doesn't come close to accounting for the $500\times$ difference in genome size.**

$$2 \times 10^4 \text{ genes} \times 375 \text{ AA/gene} \times \frac{3 \text{ AA}}{1 \text{ bp}} \approx 2.3 \times 10^7 \text{ bp.}$$

only ≈ 5 times the *E. coli* genome size.

Why is the human genome larger? **What takes up the extra space?** What do you think?

Some options include

1. transposons – jumping genes
2. *introns*
3. enhancers / regulatory sequence
4. also larger promoters
5. functional RNAs
6. maybe unknown proteins?

Interesting to note that while the number of *protein coding genes* in the human genome is $\approx 2 \times 10^4$ the estimated number of total genes is $\approx 6 \times 10^4$ including non-coding RNAs.

- Transposable elements account for $\sim 50\%$ of the human genome.
- Exons are $\sim 20\times$ shorter than introns, so the coding fraction is roughly $\frac{1}{2} \times \frac{1}{20} = \frac{1}{40} \approx \frac{1}{100}$.

Another way of thinking about introns is to think of genes as being longer. While post-splicing coding sequences are similar to bacteria in size, **human genes are much longer**, with an average length of $\approx 6,000$ bp ([Nurk et al. Science 2022.](#))

7 How big are genomes?

Figure 5 gives the range of observed genome sizes for different groups.

¹Eukaryotic proteins can be bigger, but also smaller than this. Review the vignette "[how big is the average gene?](#)" for context. It's also worth remembering various small proteins you might know about, like the antibody light chain (IgL ≈ 220 AA), various ribosomal proteins, ubiquitin, neuropeptides, and hormones. Some of these are made from larger polypeptides (ubiquitin) and some are single gene products (IgL).

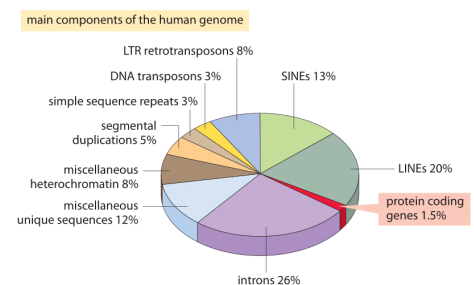


Figure 4: Approximate composition of the human genome. From Cell Biology BTN.

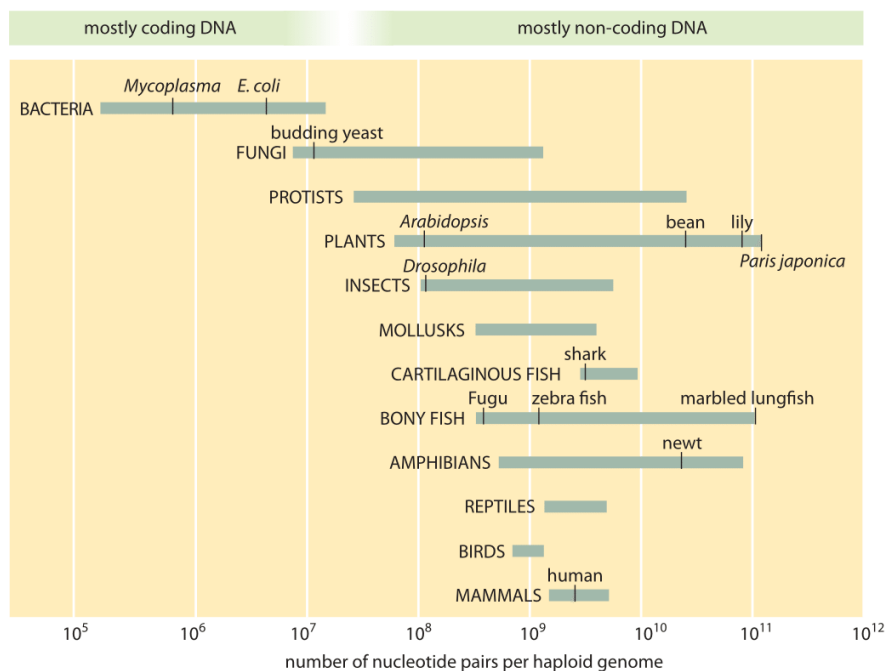


Figure 5: The sizes of various genomes. From Cell Biology BTN.

While bacterial genomes are compact (see Figure 2). There's even a protist – a non-fungal eukaryotic microbe – called *Polychaos dubium* that has a ≈ 700 Gb genome: **230× larger than the human genome**.

8 Replication time for long genomes

Consider *Polychaos dubium*. This crazy protist has a genome $\approx 200,000 = 2 \times 10^5$ fold larger than *E. coli*.

Does *Polychaos dubium* have a 10^5 hour cell cycle (≈ 10 yr) as a result?

To belabor the point, calculate the replication time from typical DNA polymerase speeds

$$f 00 \times 10^9 \text{ bp/genome} \times \frac{1}{10^3 \text{ bp/s}} \approx f \times 10^8 \text{ s}$$

$\approx \pi \times 10^7 \text{ s/yr}$, so this is ≈ 10 years!

How is this solved? What do you think?

1. genome is packaged in chromosomes (**parallelize**; $\approx 10\times$ faster)
2. slower cell cycles (≈ 1 day for many cell types)
3. **faster polymerase** ?! – the human one is actually much slower
4. multi-origin replication: another type of parallelization.

Ideas include: breaking the genome into multiple chromosomes, slower replication time, faster polymerase, or other mechanisms. This can be explored in a quantitative framework, which adds depth and allows us to assess what factors are dominant or negligible. This is really the essence of the whole course.

8.1 Do the numbers pan out for human cells

Recent work in human cells used various sequencing approaches to find origins of replication in different cell types.

A recent [synthesis by Tian et al. Elife 2024](#)) found that $f \times 10^4$ replication origins reproducibly fire across cell types (not including cancer). These are 10^5 bp apart on average. Individual segments are replicated and stitched together.

Chromosome 1 is the largest human chromosome (250×10^6 bp) — **how long would it take to replicate with this approach?**

The **trick of parallelization** is that the timing doesn't depend on the chromosome length. It **depends only on the spacing** – the distance between origins.

$$\approx 10^5 \text{ bp/segment} \times \frac{1}{10^3 \text{ bp/s}} \approx 100 \text{ s/segment}$$

Even if stitching the segments together is quite slow (say ≈ 1 hr), this is still fast enough.

8.2 Mention the end-replication problem?

See this nice introduction by [Wellinger \(2014\)](#), which summarizes of recent work in yeast ([Soudet et al., Mol Cell 2014](#)).

9 Mutations in *E. coli*

Q: You start an overnight culture from one cell. In the morning you sequence a single cell (somehow). How many mutations on average did it accumulate?

- (a) 1000 (b) 100 (c) 10 (d) 1 (e) $\ll 1$

9.1 A simpler question: mutations/replication

Q: How many mutations occur per replication of the *E. coli* genome?

- (a) 100 (b) 10 (c) 1 (d) $\ll 1$

Anyone know something about replication error rates? Turns out that it's not just about the polymerase. You can think about the (i) replication fidelity, (ii) proofreading activity by exonuclease domains, (iii) mismatch repair by other factors, and (iv) the DNA damage response. The contribution of these factors has been dissected piece by piece in *E. coli* ([Niccum et al. Genetics 2018](#)).

This approach, as well as analysis of mutation accumulation in the Lenski lines (Fig. 6), gives a replication error rate of 10^{-10} errors/base/gen.

9.2 Calculation

- Replication error rate (after proofreading): 10^{-10} errors/b/gen
- Genome length: 5×10^6 bp $\Rightarrow 10^7$ nucleotides added per replication
- Mutations per replication: $10^{-10} \times 10^7 \approx 10^{-3}$ **(answer: d)**
- Overnight: $\sim 10^{10}$ cells, ~ 30 replications per lineage

$$2 \text{ b/bp} \times 5 \times 10^6 \text{ bp/genome} \times 10^{-10} \text{ errors/b/gen.} \approx 10^{-3} \text{ mutations/gen}$$

$$30 \text{ gen} \times 10^{-3} \text{ mutations/gen} \approx f \times 10^{-2} \text{ mutations/progeny}$$

$\Rightarrow$ After an overnight, a single cell is essentially identical to its ancestor.

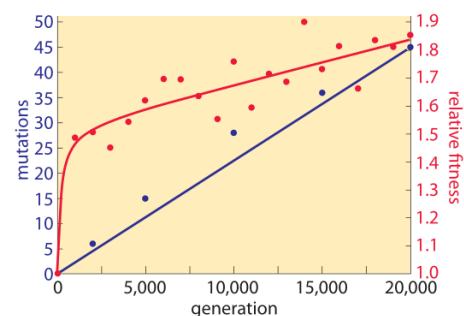


Figure 6: Mutation accumulation and fitness over time in Richard Lenski's 30 yr *E. coli* evolution experiments. Sequencing measurements make it possible to examine the rate of mutation accumulation and the corresponding fitness over time. Adapted from J. E. Barrick et al. *Nature*, 461:1243, 2009. Figure from Cell Biology BTN.

9.3 Analogy to Errors in Books and Data Storage

A book with 50 letters/line and 50 lines/page has 2500 characters/page.

- 10^{-4} error rate: one error in ~ 4 pages.
- 10^{-10} error rate: one error in $\sim 4,000,000$ pages — more than a 1000-volume encyclopedia.
- Modern data storage achieves 10^{-14} : approximately 1 error per 12 TB.

10 SARS-COV-2 Alpha variant

The Alpha variant (B.1.1.7) was [3][4] a SARS-CoV-2 variant of concern. It was estimated to be 40–80% more transmissible than the wild-type SARS-CoV-2 (with most estimates occupying the middle to higher end of this range). Scientists more widely took note of this variant in early December 2020, when a phylogenetic tree showing viral sequences from Kent, United Kingdom looked unusual.

– [Wikipedia](#)

Does anyone remember how many mutations Alpha had relative to the Wuhan strain?

Alpha had ≈ 30 mutations relative to Wuhan. If you remember, **this seemed like a lot of mutations at the time.**

Is it plausible that this arose by evolution over ≈ 10 months.² Was it the result of evolution?

Two approaches: **microscopic** (mechanistic) and **macroscopic**/empirical.

10.1 Microscopic / mechanistic approach

Mechanistic approach – think about the polymerase, error rate, number of viral replication cycles.

$$E[N_{\text{mut}}] = \text{mutation rate} \times \text{genome size} \times \text{num. replications}$$

We know the genome size and have reasonable estimates for some viral RNA polymerases³, **but this approach poses some challenges.** First, **we don't know the number of viral replication cycles.** Second, **the polymerase error rate can evolve** along with other aspects of the virus.

For these reasons, we prefer the macroscopic approach.

10.2 Macroscopic estimate

I definitely gave my wife COVID. If we sequenced my and her virions, how many mutations would we have found?

The idea here is to **think about the infection process** instead of the molecular mechanism of replication. So we **count infection cycles instead of viral generations.**

$$E[N_{\text{mut}}] = \frac{\# \text{ mutations}}{\text{infection cycle}} \times \frac{\text{time elapsed } (\approx 1\text{yr})}{\text{infection cycle duration } (\approx 5 \text{ days})}$$

²First confirmed case in Wuhan was in Dec. 2019.

³Coronaviruses have RNA genomes.

- Mutation rate: ≈ 0.5 mutations per infection cycle (easier to measure — sequence two people).
- Note the difference between one lineage and all progeny.
- Infection cycle length: ≈ 5 days
- Rate: ~ 3 mutations/month
- Time to accumulate ~ 30 mutations: ≈ 10 months

COVID began $\approx$ December 2019. Alpha appeared $\approx$ December 2020 (Hanukkah). That is ≈ 10 months — consistent!

We don't need special explanations e.g., genetic engineering or infections of immunosuppressed people.

11 Error rate in protein translational

Q: What fraction of proteins contain at least one translational error?

- (a) 1 in a million (b) 1 in a thousand (c) 1% (d) $> 1\%$

11.1 Calculation

- Translation error rate: 10^{-3} – 10^{-4} errors/codon
- Fraction of proteins with at least one error: $1 - (1 - p)^n$, where $n \approx 300$ codons

$$1 - (1 - 10^{-4})^{300} \approx 3\%$$

$$1 - (1 - 10^{-3})^{300} \approx 26\% \quad (\approx 30\%)$$

Answer: (d) $> 1\%$

11.2 Transcription error rate:

RNA polymerase error rates are 10^{-4} – 10^{-5} errors/nt. For a 1000 nt transcript:

$$1 - (1 - 10^{-4})^{1000} \approx 10\%, \quad 1 - (1 - 10^{-5})^{1000} \approx 1\%$$

(Note: error-prone viruses also operate in the 10^{-4} – 10^{-5} range.)

11.3 Analogy: digitizing the Oxford English Dictionary

As an example, during the transfer of the printed version of the Oxford English Dictionary to electronic form ('keyboarding') the error rate was about 1 in 2000 characters over the 350 million total even with the large element of self-checking by the internal consistency in spelling.

—(R. S. Clymo)

11.4 Analogy to errors in books and data storage

A book with 50 letters/line and 50 lines/page:

- 10^{-4} error rate: one error in ~ 4 pages.
- 10^{-10} error rate: one error in $\sim 4,000,000$ pages — more than a 1000-volume encyclopedia.
- Modern data storage achieves 10^{-14} : approximately 1 error per 12 TB.

Solution: Quality control, premature translation termination, and robustness.

12 Specific mutation Already Present in a Culture?

Question for recess:

Q: You need a specific base-pair mutation (G→A at position 2,918,483) in at least one cell of a 5 mL bacterial culture. You need to perform

- (a) Random mutagenesis (e.g. UV/EMS);
- (b) Site-directed mutagenesis;
- (c) do nothing.

12.1 Solution

Overnight culture: 5 mL, OD $\approx$ 2, 10^9 cells/mL/OD $\Rightarrow$ 10^{10} cells.

$$10^{-10} \text{ errors/base/rep} \times 10^{10} \text{ cells} \approx 1$$

$\Rightarrow$ The mutation is **already there**. (A factor of 3 applies since any of 3 bases could substitute, but the conclusion stands.)

12.2 Alternative Derivation

$$10^{-3} \text{ mutations/rep.} \times 10^{10} \text{ recent replications} \approx 10^7 \text{ mutations}$$

Compare to $f \times 10^6$ bp in a bacteria genome. The genome is **saturated** — every single-base mutation has occurred.

13 De Novo Mutations in Humans

Q: If we are looking for a newly arising specific mutation in the global human population, will we find it? (E.g. a specific de novo mutation conferring resistance to Ebola.)

$$10^{-10} \text{ mutations/bp/replication} \times 10^{10} \text{ bp/genome} \approx 1 \text{ mutation/replication}$$

Thoughts about this? Anything we need to discuss

1. conserved regions – mutations may not be tolerated, lose these.
2. really care about human generations, not genome replications!

Important to notice: replication $\neq$ generation. Somatic mutations are irrelevant; only germline (gamete) replications count for passage to the next generation.

13.1 Replications per human generation

Values differ between mother and father. Typical values are:

- Female: $\sim$ 23 replications/generation (range 20–30)
- Male: $\sim$ 35 replications/generation at age 15; $\sim$ 800/generation at age 50 (range 40–800)

$\Rightarrow$ **Most mutations arise in the male germline.**

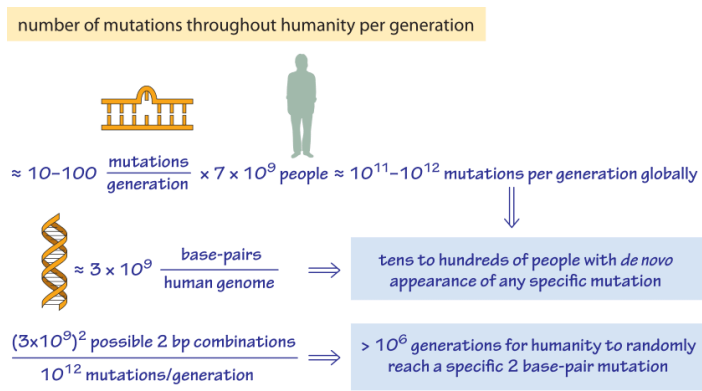


Figure 7: Estimating the number of mutations formed in all of humanity. From Cell Biology BTN.

13.2 New human mutations per generation

Take 100 replications/generation as a typical number based on the paternal side. This gives.

$$1 \frac{\text{mutations}}{\text{genome replication}} \times 10^2 \frac{\text{replications}}{\text{generation}} \approx 10^2 \frac{\text{mutations}}{\text{human generation}}$$

$$10^2 \frac{\text{mut}}{\text{gen}} \times 10^{10} \text{ people} = 10^{12} \text{ new mutations per generation}$$

Comparing 10^{12} mutations to a human population of $8 \times 10^9 \approx 10^{10}$ means that **each mutation appears in tens to hundreds of people** (Fig. 7).

If we compare the total number of mutated bp ($\approx 10^{12}$) to the total number of bp in all people (population $\times$ genome size)

$$\approx \frac{10^{12} \text{ mutations/generation}}{10^{10} \text{ people} \times f \times 10^9 \text{ bp/genome}} \approx \frac{1 \text{ mutated bp}}{f \times 10^7 \text{ total bp} \times \text{generation}}$$

So we estimate per-site mutation rate of $f \times 10^{-8}$ mutations/site/generation, within a factor of f of empirical values [BNID 112459].

14 Double mutations in humans

Q: You want a specific *de novo* double mutation (G→A at position 2,918,483,453 and T→C at position 2,918,425,291) in at least one living person.

One way of phrasing: what are the odds of finding it in the human population?

Another way of phrasing: how long do I need to wait to find this mutation? First mutation present in 10 – 100 people. Then I need the second mutation to arise in one of these people.

14.1 Calculation

The first mutation is present in ≈ 100 people. The probability of any one of those individuals also acquiring the second mutation is

$$p_2 \approx 100 \text{ ppl} \times f \times 10^{-8} \text{ mutations/site/gen.} \times 1 \text{ sites/person} \approx f \times 10^{-6} \text{ mut/person/gen}$$

If the frequency of the first allele doesn't change much, i.e., remains near 100 people, then we expect to wait

$$\frac{1}{f \times 10^{-6}} \approx f \times 10^5 \text{ generations}$$

Which is a few million years.

Caveat

14.2 Conclusions

- One specific base substitution: easy (already present in large populations).
- Two specific base substitutions: very hard. This holds for bacteria as well.
- **Combinatorics is tough.**
- Selection is critical; neutral drift creates diversity.
- The benefits of sexual reproduction, and other mechanisms (insertions, transposable elements, etc.) become clear in this framework.
- Each mutation must be beneficial, or there must be substantial neutral variation, or other deep evolutionary principles must apply.
- Quantitative thinking helps identify what is dominant and what is negligible.
- Despite all of that – we evolved from very simple things like bacteria.

Thoughts?

1. Random vs. non-random effects
2. Sex / gene transfer – mixing useful variants
3. Cancer – often said it requires multiple mutations

We did all our calculations for single nucleotide polymorphisms (SNPs), but **it's not only SNPs that matter** – other kinds of mutations. What is there?

1. Duplication
2. Rearrangements
3. Foreign genes - HGT, endosymbiosis
4. Splicing
5. Mention neutral drift

Often when you do evolution in the lab you see that these other mechanisms matter a lot. For example, yeast and E. coli perform large chromosomal rearrangements during laboratory evolution (see [Yona et al. PNAS 2012](#), [Raeside et al. MBio 2014](#)).